VF Asia Footwear Laboratory Testing Manual

Test Method for Coefficient of Friction- TM144 (Traction Properties)	Test method # VF- B25 REF # : TM-R- 066 Page 1 of 5
Date Issued : July 30, 2014 Revision # :	Created by : Nila Babia Erazo / Chris Adam

PURPOSE AND SCOPE: To evaluate the traction properties of VF footwear and outsole compounds.

REFERENCES: SATRA Test Method TM 144, EN 13287:2004 ISO 13287:2004

EQUIPMENT AND MATERIAL:

1. SATRA STM-603 Slip Resistance Tester or equivalent meeting the requirement of SATRA 144

2. Surface contaminants.
   - Distilled water
   - Oily Wet #1 – 2 drops (~0.025g per drop) of vegetable oil per 25mL distilled water
   - Oily Wet #2 – 7 drops (~0.025g per drop) of vegetable oil per 25mL distilled water
   - Oily Wet #3 – 8 drops (~0.025g per drop) of vegetable oil per 25mL distilled water
   - NaLS – 5% w/w sodium lauryl sulphate, general laboratory grade, in distilled water

3. Test surfaces :
   - The standard surface shall be SATRA Clay Quarry Tile (Ref. 1.1 section B3.1)
   - Alternate surfaces :
     - Stainless Steel (Ref. 1.1 section B3.2)
     - Frosted Rough Ice
     - Boat decking/glass reinforced plastic

4. Material Slider 96’ as defined in SATRA TM 144.
5. 76 ± 1 mm (3”) diameter die.
6. A means of attaching a die-cut test specimen (slider) to the test machine. A description can be found in SATRA TM144:2007, Annex B, section B5.1.5.
7. Footwear to be tested.
8. Test compounds / molded test plaques :
   - When testing compounds, molded test plaques should be made using the LAB-1779 mold. The tread pattern on the molded part should conform with the design in Figure 1. The tread pattern shall be flat and free of texture. The mold to be used for making test plaques can be obtained from the following supplier:

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Jye Shing Precision Mold Co.,Ltd
Taiwan mother company:
Address: 48 Lane, Sec3,Chung Shan Rd.,Taiping Taichung. Taiwan
Tel:886-4-22705745 FAX 886-4-22790124,22760838
E-mail:jsyshing@ms8.hinet.net
Mainland company
Address: Xia Jiang Chong Industrial, Gao Bu Town, Dong Guan, Guang Dong, China
Tel:86-769-88871347,88872347, FAX:86-769-88870346,88871346
E-mail:jyeshing@jyeshing.com.tw

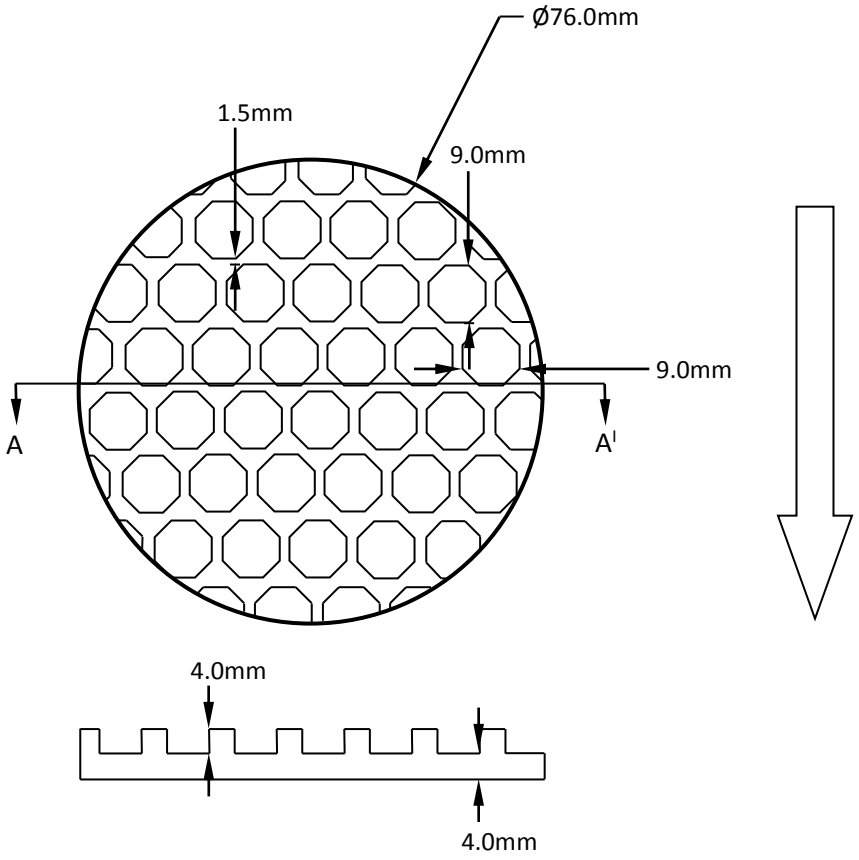


Figure 1. Tread pattern design (LAB-1779) for testing outsole compounds.

Arrow denotes test direction.

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SAMPLE PREPARATION

Condition the specimens at 23 ± 2 ℃ temperature and 55 ± 10% relative humidity for 24 hours prior to test.

REPETITIONS: Perform three (3) test / sample.

TEST PROCEDURE:

A. <u>Whole Shoe Testing :</u>

1. Footwear shall be prepared and tested in accordance with SATRA TM144: 2007. Exceptions to this procedure are listed below:

   • Footwear should be tested as received, without cleaning.

   • Footwear should be tested after cleaning, as outlined in SATRA TM144:2007, Annex D, section D2.

   • Any discrepancy in results should be reported.

2. Standard Quarry Tile shall be used for the testing surface. Unless otherwise specified. Alternate testing surfaces could include stainless steel, rough ice, or glass reinforced plastic.

3. Testing shall be conducted in both the heel and forefoot, in accordance with the procedures outlined in SATRA TM144: 2007.

   • The heel shall be tested at an angle of 7 ± 0.5° relative to the testing surface. (Ref 1.1 section A3.4)

   • The forepart shall be tested at an appropriate angle with respect to the test surface and so that the joint of flexing area of the footwear forepart is in contact with the floor. (Ref 1.1 section A3.5)

   • The speed of the test shall be 30 cm/sec. (Ref 1.1 section 4.8)

4. Footwear shall be tested dry and using distilled water as a surface contaminant.

5. When surface contaminants are used, the test surface should be wetted completely with the contaminant prior to starting the test. Complete wetting is indicated by a puddle of the contaminant in the area where the shoe will contact the test surface. The contaminant shall be reapplied between tests as necessary to keep the test surface wetted.

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6. PRO product, products to be marketed as “Anti-slip”, products marketed with special performance properties, or products labeled with compliance to a specific performance specification (ANSI, ISO, EN, CE) may require testing with additional contaminants or test surfaces. These contaminants or surfaces will be specified at the time of testing and might include:

    • 0.5% NaLS solution.

    • Oily Wet #1, #2, or #3. When using these contaminants, the oil drops should be rubbed into the test surface first. Water should then be applied prior to testing.

    • Frosted rough ice.

B. Outsole Compound Testing

1. The molded test parts should be die cut using the 76 ± 1 mm (3”) diameter die. Testing should be conducted in the direction outlined in Figure 1.
2. Standard Quarry Tile shall be used for the testing surface.
3. The molded test part should be tested with the disc set at an angle of 7 ± 0.5° relative to the testing surface (Ref. 1.1 section A3.4)
4. The speed of the test shall be 30 cm/sec (Ref. 1.1 section 4.8)
5. The molded test part shall be tested dry and using distilled water as a surface contaminant. Certain specialized compounds on the Timberland Rubber Matrix (Ref 1.4) also require the use of NaLS as a surface contaminant.
6. When surface contaminants are used, the test surface should be wetted completely with the contaminant prior to starting the test. Complete wetting is indicated by a puddle of the contaminant in the area where the test plaque will contact the test surface. The contaminant shall be reapplied between tests as necessary to keep the test surface wetted.
7. Compounds used for PRO product, products to be marketed as “Anti-slip”, products marketed with special performance properties, or products labeled with compliance to a specific performance specification (ANSI, ISO, EN, CE) may require testing with additional contaminants or test surfaces. These contaminants or surfaces will be specified at the time of testing and might include:

    • Oily Wet #1, #2, or #3. When using these contaminants, the oil drops should be rubbed into the test surface first. Water should then be applied prior to testing.

    • Frosted rough ice.

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CALIBRATION:

1. Calibration shall be performed at regular intervals, using the standard 'Slider 96,' in accordance with SATRA TM144: 2007, which states:

   "Calibration of the tiles should be checked after every 10 tests or prior to each day of testing whichever is the less frequent, to ensure that they are not being worn smooth or otherwise damaged."

2. When calibrating, quarry tiles shall fall within the following ranges as outlined by SATRA:

Surface Condition	Calibration Range
Dry	0.57 – 0.63
Wet	0.36 – 0.42

REPORT:

1. The coefficient of friction for each mode of the test : COF

2. Should also include the following :
   - A reference to the test method used: SATRA TM144:2007.
   - A full description of the footwear or outsole compound
   - A description of the test surface.
   - A description of surface contaminants.
   - A description of how the footwear or outsole compound was tested: Heel, Forefoot, or Flat.